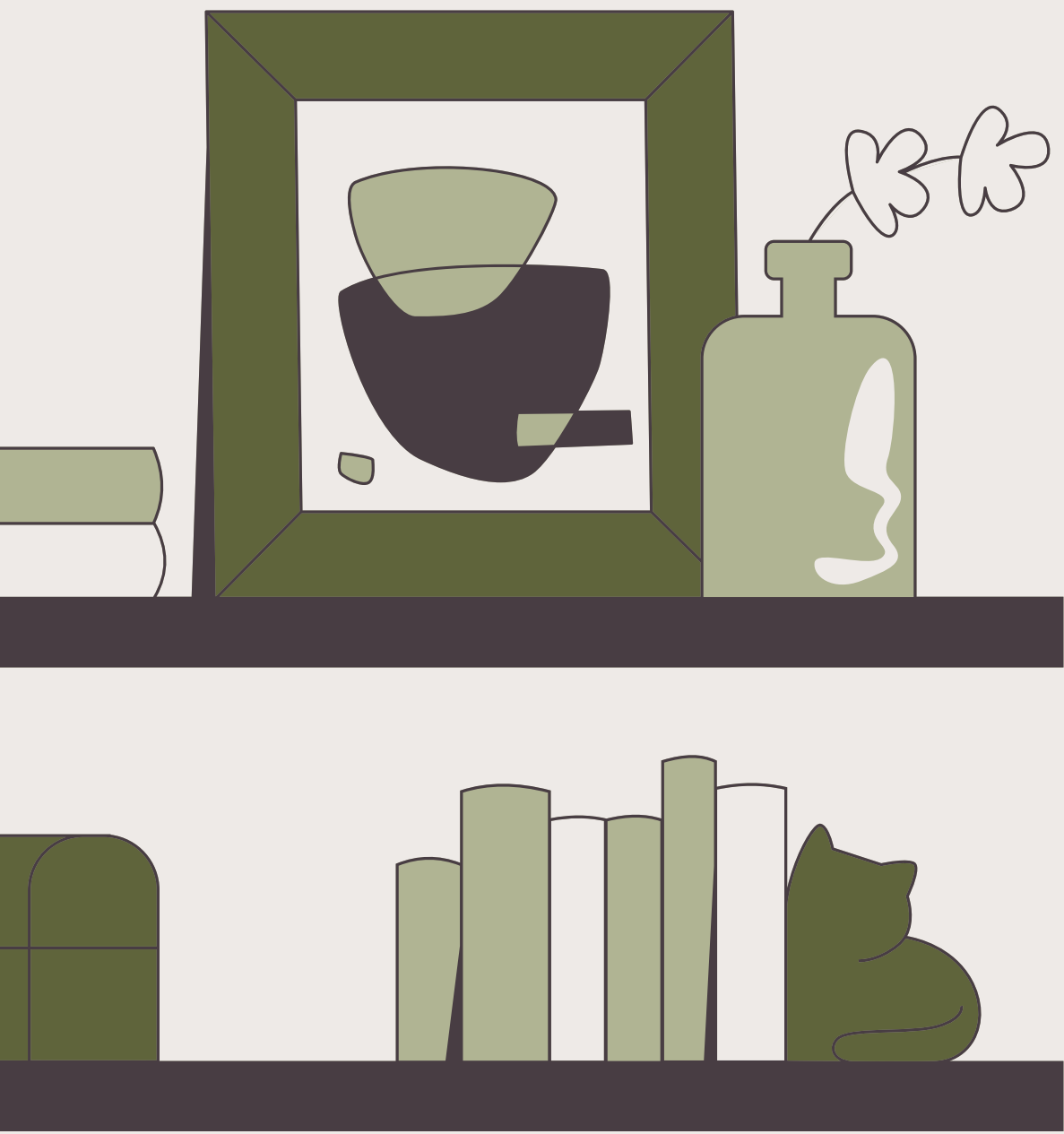




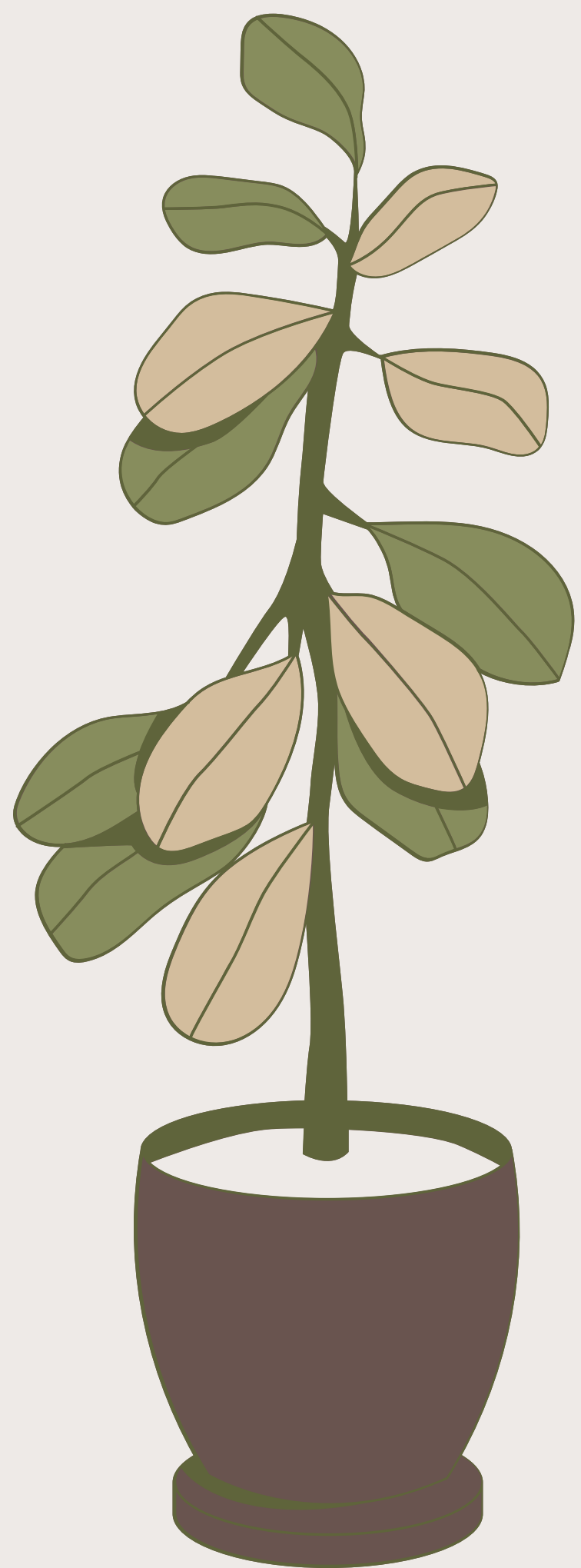
# SQL Project: Bookstore Management System





## Tools Used

- *PostgreSQL – Database Management*
- *PgAdmin – Query Execution & Database Handling*
- *Microsoft Excel / CSV Files – Data Import*
- *Canva – Project Presentation Design*
- *SQL – Data Querying & Analysis*







# *Database Schema (Books, Customers, Orders)*

## **BOOKS**

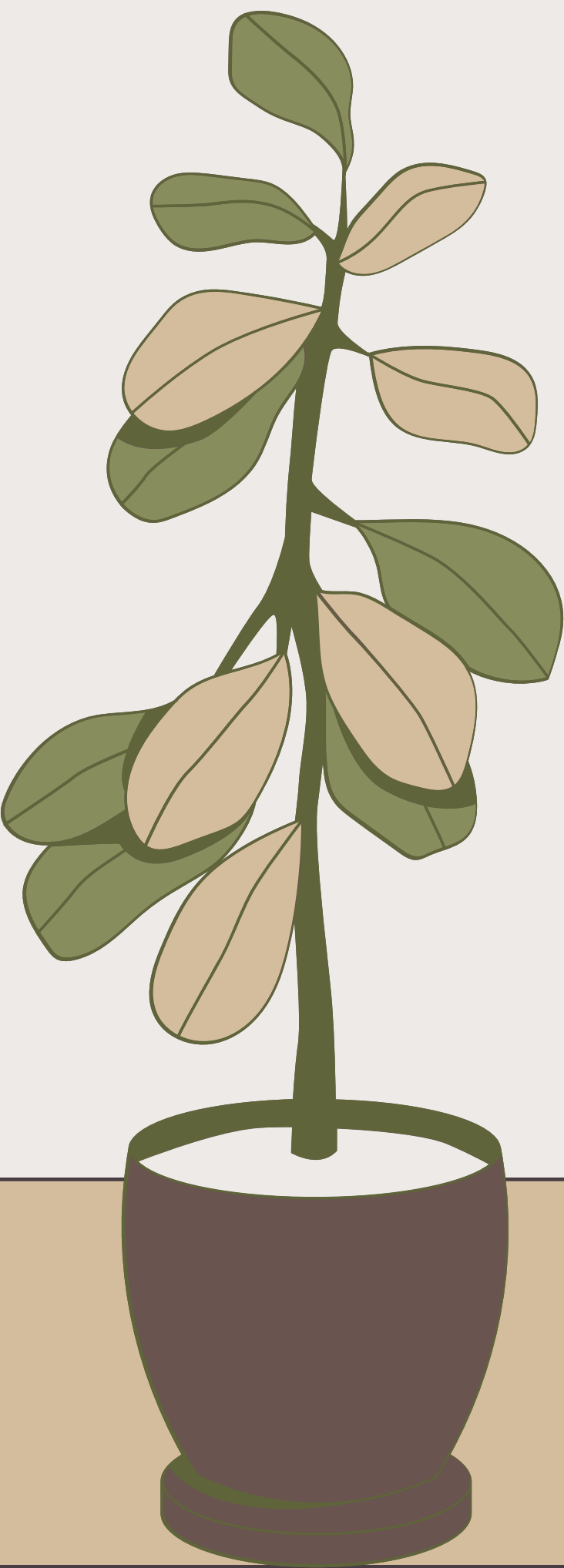
- Book\_ID (PK)
- Title
- Author
- Genre
- Published\_Year
- Price
- Stock

## **CUSTOMERS**

- Customer\_ID (PK)
- Name
- Email
- Phone
- City
- Country

## **ORDERS**

- Order\_ID (PK)
- Customer\_ID (FK)
- Book\_ID (FK)
- Order\_Date
- Quantity
- Total\_Amount



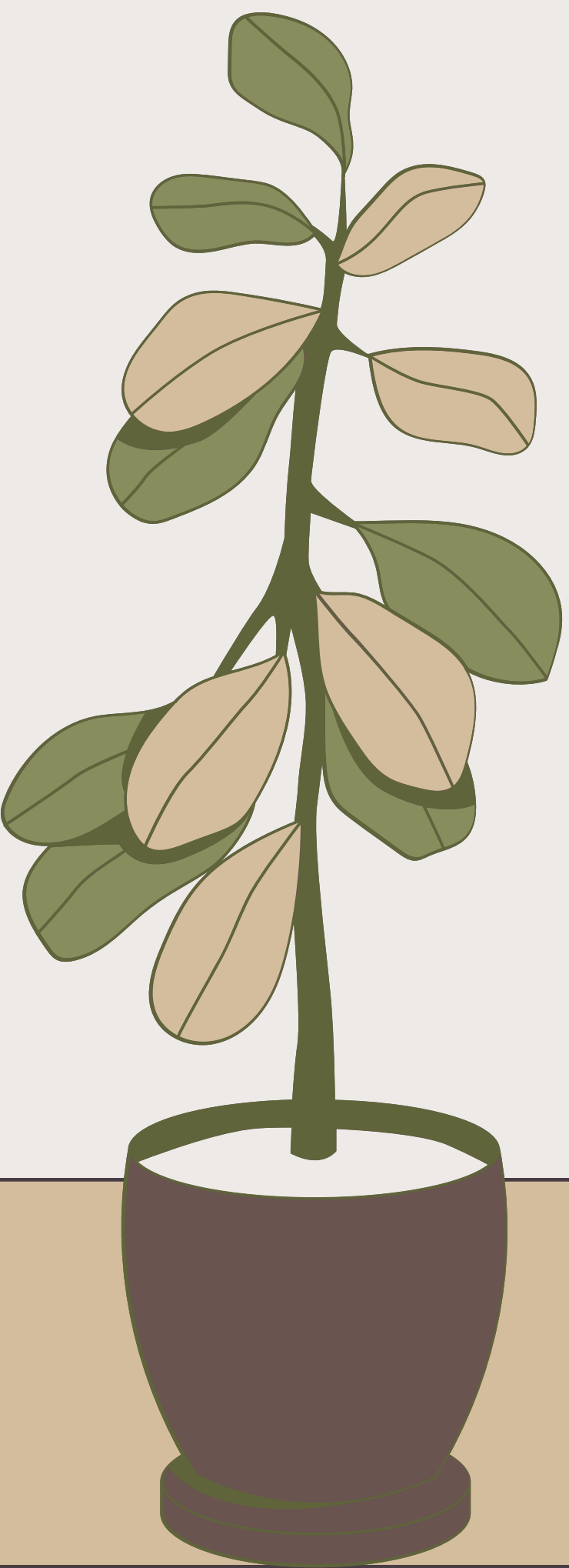




## ***Creating & importing Books Table***

```
--Creating Books Table
Create table books(
Book_ID serial primary key,
Title varchar(100),
Author varchar(100),
Genre varchar(50),
Published_Year int,
Price Numeric(10,2),
Stock int
);

--Import data into table
COPY books(Book_ID, Title, Author, Genre, Published_Year, Price, Stock)
FROM 'D:\POSTERES_SQL\ST - SQL ALL PRACTICE FILES-2\Project_PgSql\Books.csv'
CSV HEADER;
```







## ***Creating & importing Customer Table***

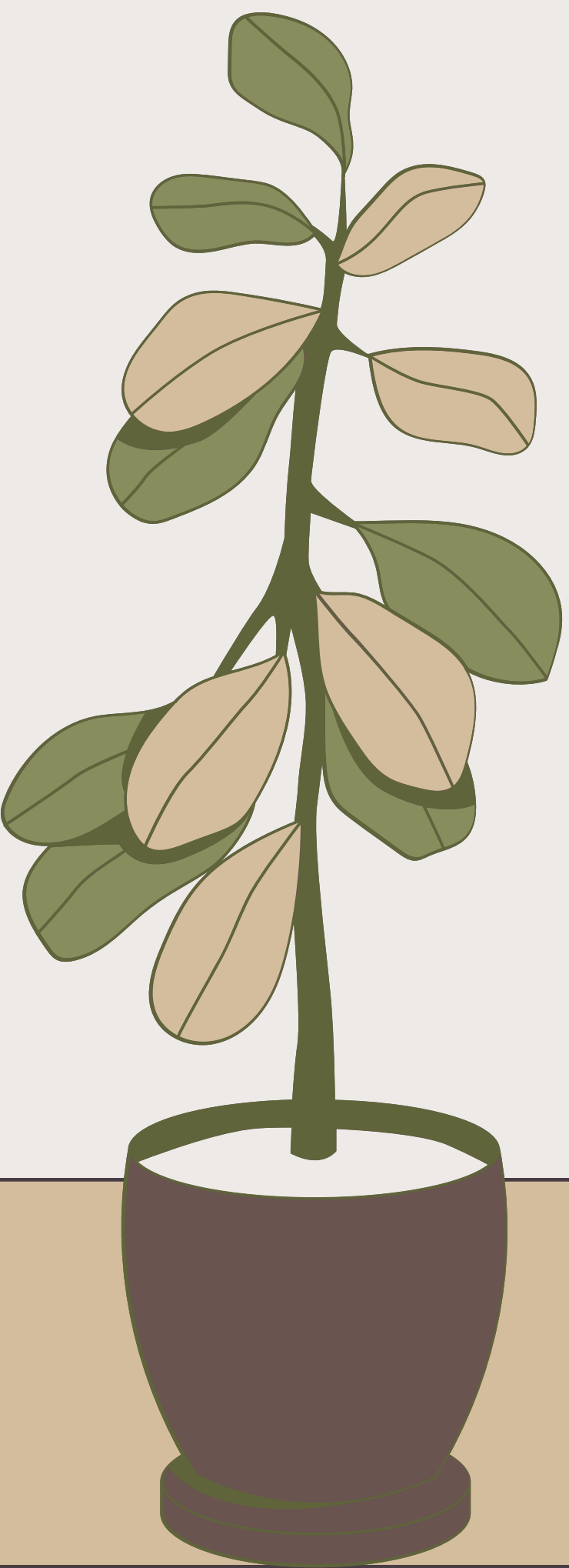
```
--Creating Customer Table
Create table Customers(
Customer_ID serial primary key,
Name Varchar(100),
Email Varchar(100),
Phone Varchar(15),
City Varchar(50),
Country Varchar(150)
);
--Import data into table
COPY Customers(Customer_ID, Name, Email, Phone, City, Country)
FROM 'D:\POSTERES_SQL\ST - SQL ALL PRACTICE FILES-2\Project_PgSql\Customers.csv'
CSV HEADER;
SELECT * FROM CUSTOMERS;
```





## ***Creating & importing Orders Table***

```
--Create Orders Table
Create table Orders(
Order_ID serial primary key,
Customer_ID int references Customers(Customer_ID),
Book_ID int references books(Book_ID),
Order_Date Date,
Quantity int,
Total_Amount numeric(10,2)
);
--Import data into table
COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)
FROM 'D:\POSTERES_SQL\ST - SQL ALL PRACTICE FILES-2\Project_PgSql\Orders.csv'
CSV HEADER;
select * from Orders;
```





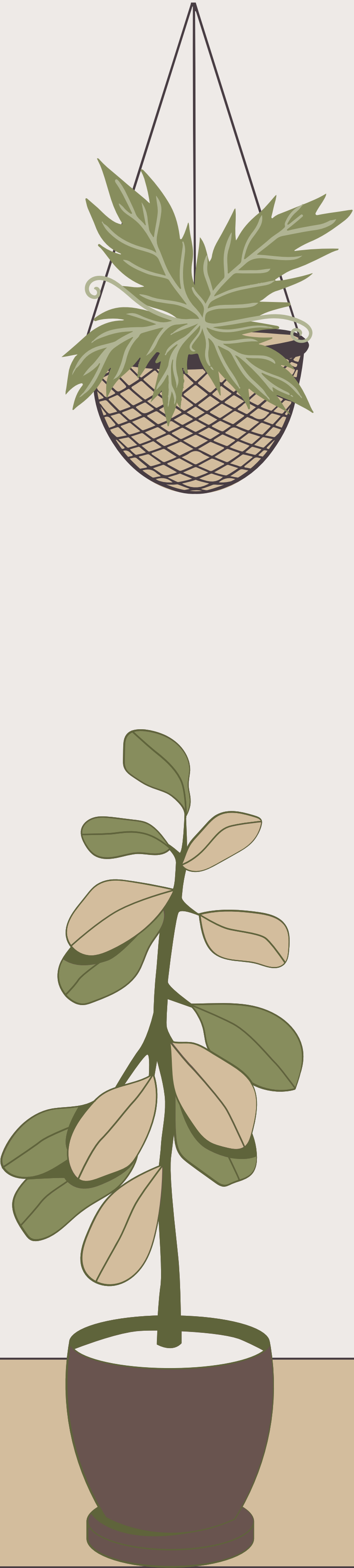


## --Questions--

--Retrive all the books in the "Fiction" Genre:

```
SELECT
    *
FROM
    BOOKS
WHERE
    GENRE = 'Fiction';
```

|   | book_id<br>[PK] integer | title<br>character varying (100)        | author<br>character varying (100) | genre<br>character varying (50) | published_year<br>integer | price<br>numeric (10,2) |
|---|-------------------------|-----------------------------------------|-----------------------------------|---------------------------------|---------------------------|-------------------------|
| 1 | 4                       | Customizable 24hour product             | Christopher Andrews               | Fiction                         | 2020                      | 43.52                   |
| 2 | 22                      | Multi-layered optimizing migration      | Wesley Escobar                    | Fiction                         | 1908                      | 39.23                   |
| 3 | 28                      | Expanded analyzing portal               | Lisa Coffey                       | Fiction                         | 1941                      | 37.51                   |
| 4 | 29                      | Quality-focused multi-tasking challenge | Katrina Underwood                 | Fiction                         | 1905                      | 31.12                   |
| 5 | 31                      | Implemented encompassing conglomerat... | Melissa Taylor                    | Fiction                         | 2010                      | 21.23                   |





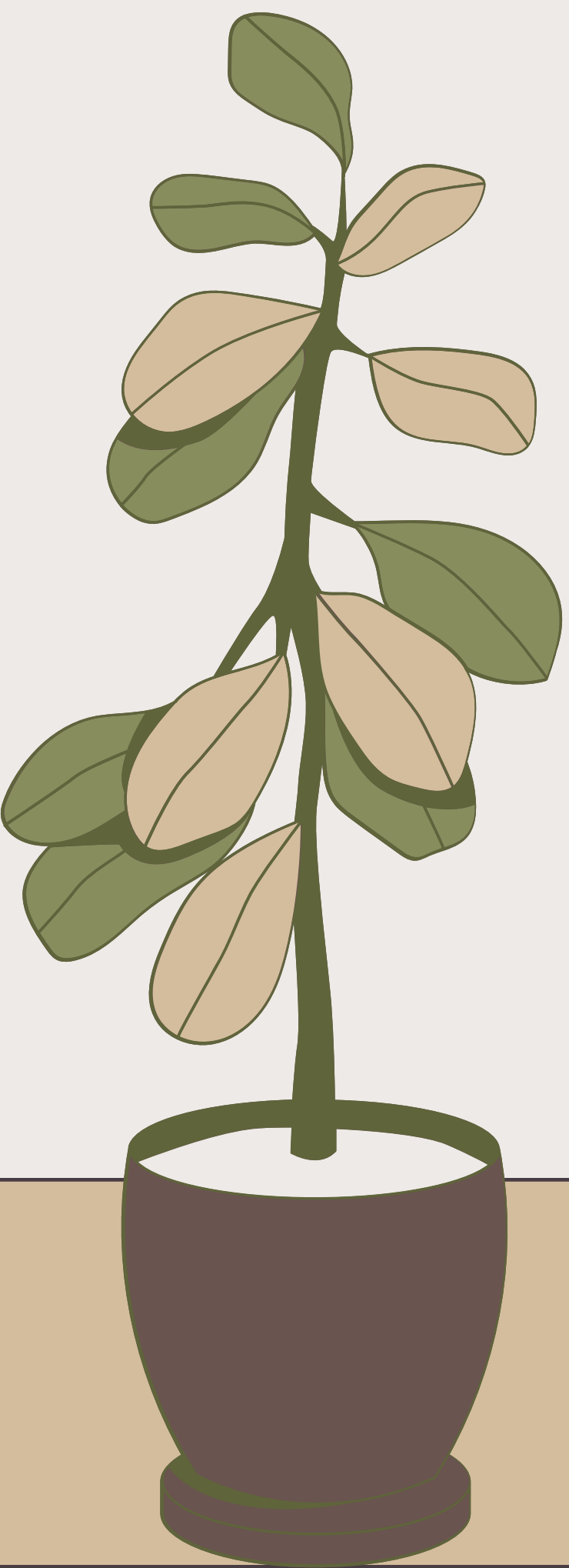


## --Questions--

--Retrive the total stock of books available:

```
SELECT
    SUM(STOCK) AS TOTAL_STOCKS
FROM
    BOOKS;
```

|   |                        |   |
|---|------------------------|---|
|   | total_stocks<br>bigint | 🔒 |
| 1 | 25056                  |   |



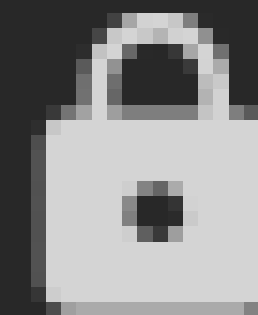




--Calculate the total revenue generated from all orders:

```
SELECT  
    SUM(TOTAL_AMOUNT) AS REVENUE  
FROM  
    ORDERS;
```

revenue  
numeric



75628.66

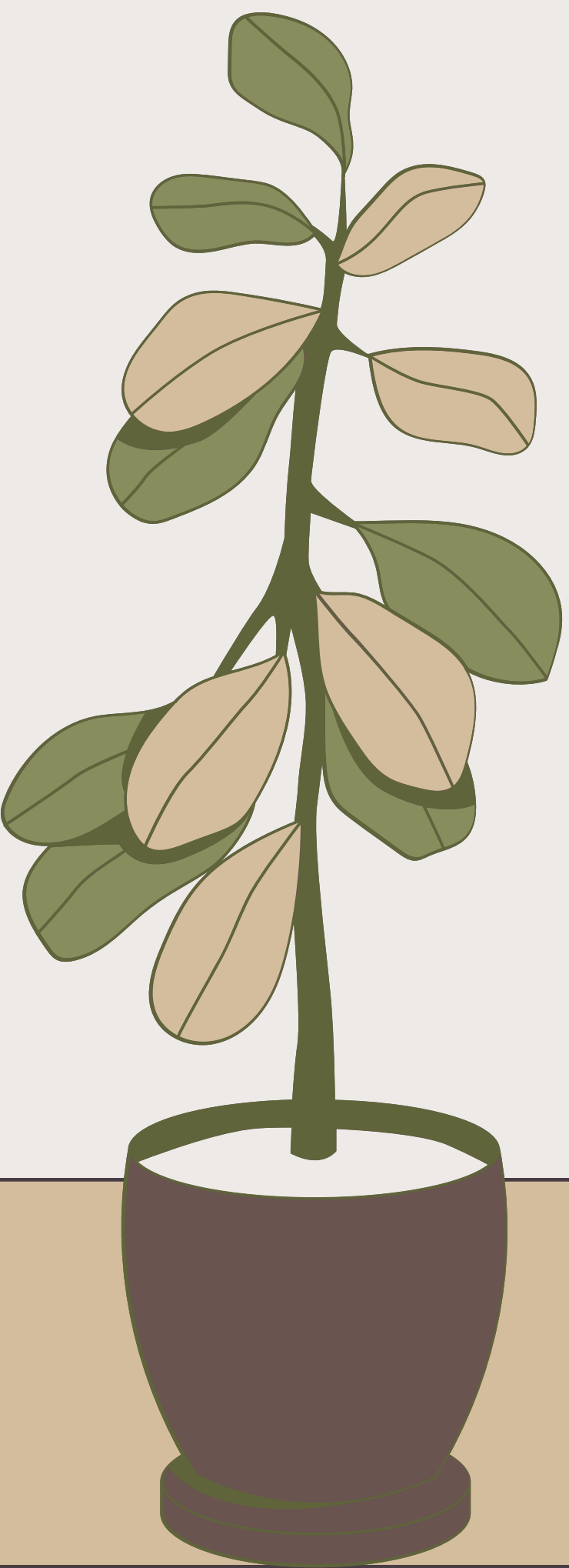






--Retrieve the total numbers of books sold for each genre:

```
SELECT
    B.GENRE,
    SUM(O.QUANTITY) AS TOTAL_BOOKS_SOLD
FROM
    ORDERS O
    JOIN BOOKS B
    ON O.BOOK_ID = B.BOOK_ID
GROUP BY
    B.GENRE;
```

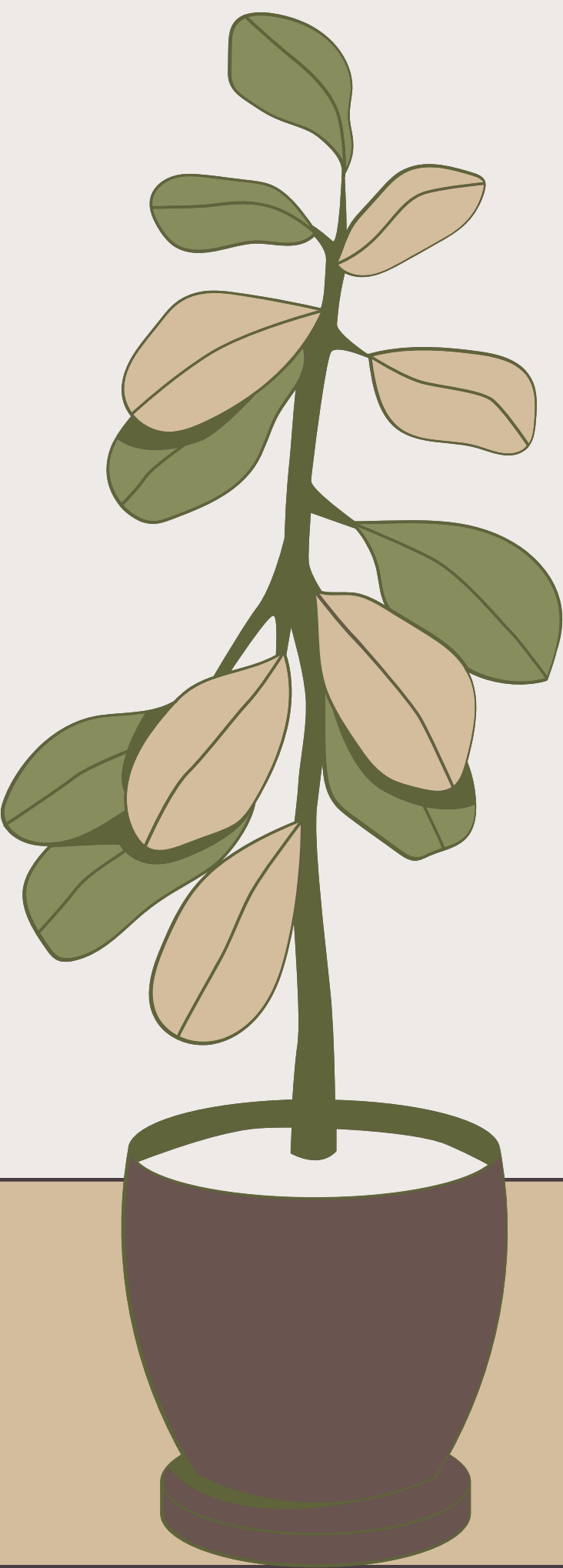






# #OUTPUT#

|   | genre<br>character varying (50)  | total_books_sold<br>bigint  |
|---|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1 | Romance                                                                                                             | 439                                                                                                            |
| 2 | Biography                                                                                                           | 285                                                                                                            |
| 3 | Mystery                                                                                                             | 504                                                                                                            |
| 4 | Fantasy                                                                                                             | 446                                                                                                            |
| 5 | Fiction                                                                                                             | 225                                                                                                            |
| 6 | Non-Fiction                                                                                                         | 351                                                                                                            |
| 7 | Science Fiction                                                                                                     | 447                                                                                                            |

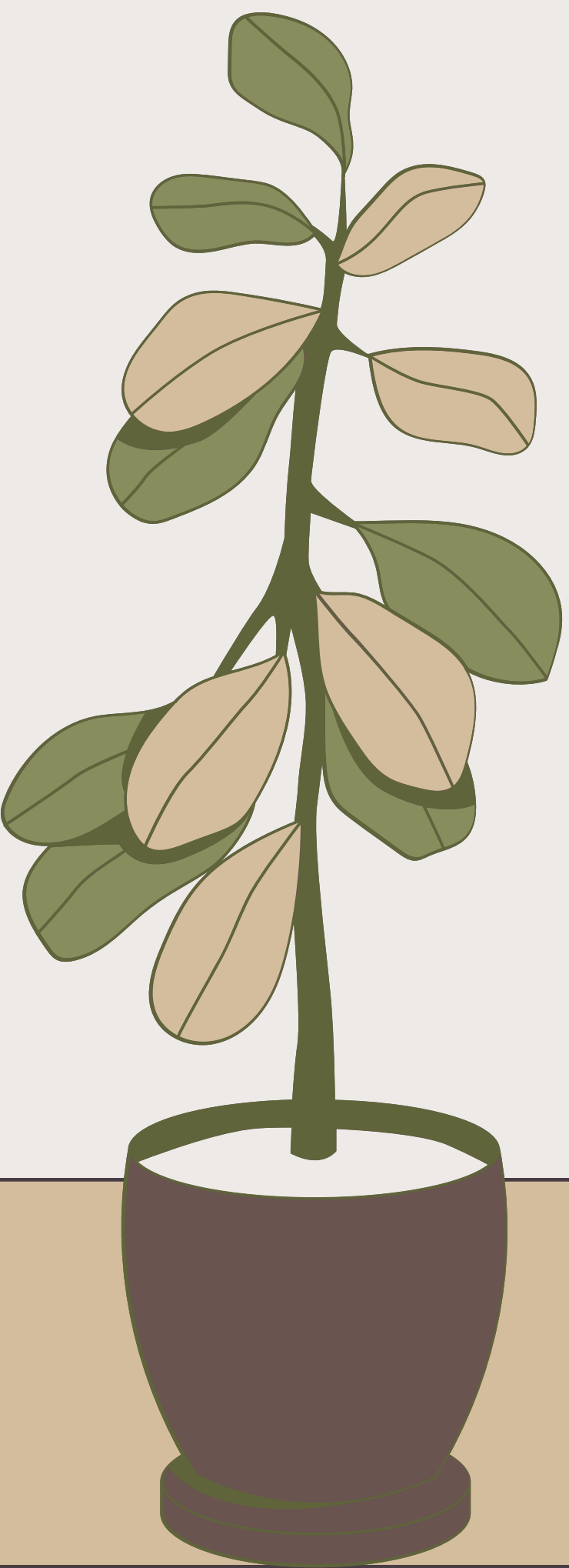






--Find the avrage price of book in fantacsy genre:

```
SELECT
    AVG(PRICE) AS AVRAGE_PRICE
FROM
    BOOKS
WHERE
    GENRE = 'Fantasy';
```



|   | avrage_price<br>numeric |
|---|-------------------------|
| 1 | 25.9816901408450704     |





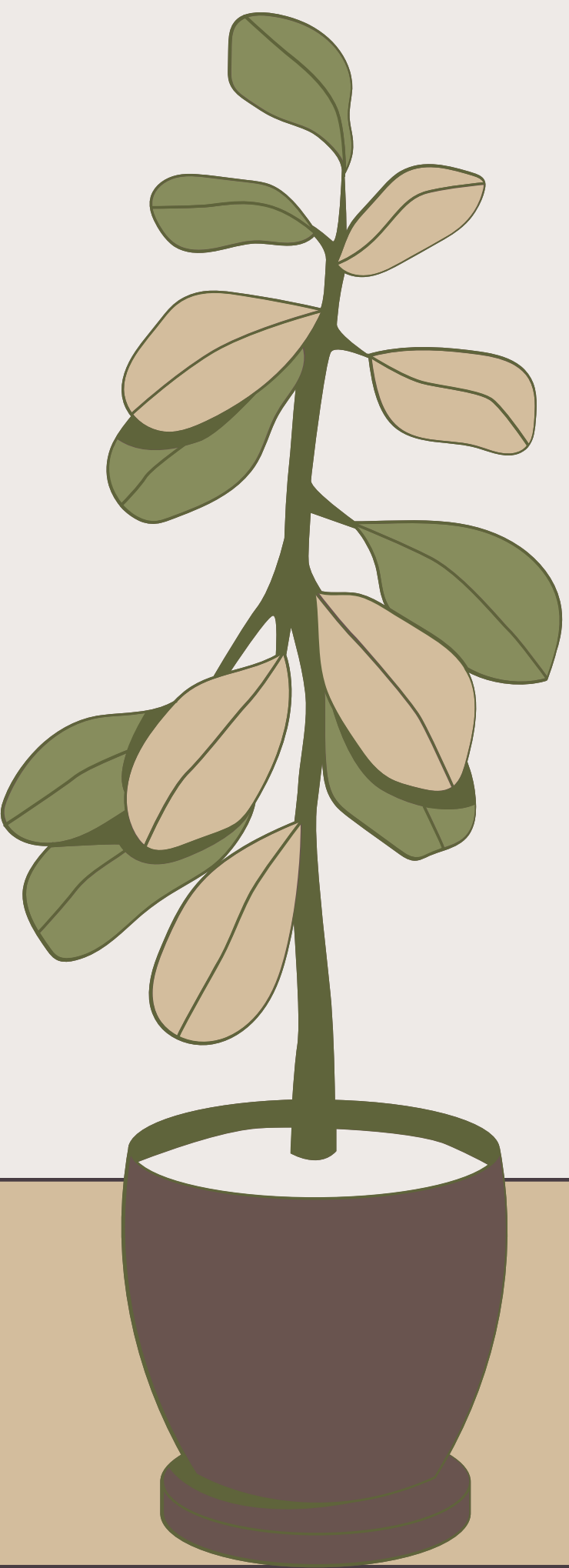


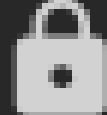
--List the customers who have placed at least 2 orders:

```
SELECT
  O.CUSTOMER_ID,
  C.NAME,
  COUNT(O.ORDER_ID) AS ORDER_COUNT
FROM
  ORDERS O
  JOIN CUSTOMERS C ON O.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY
  O.CUSTOMER_ID,
  C.NAME
HAVING
  COUNT(ORDER_ID) >= 2;
```







|  | customer_id  | name  | order_count  |
|--|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|  | integer                                                                                         | character varying (100)                                                                  | bigint                                                                                          |
|  | 225                                                                                             | Christopher Mccullough                                                                   | 2                                                                                               |
|  | 418                                                                                             | Kiara Blankenship MD                                                                     | 3                                                                                               |
|  | 322                                                                                             | William Cameron                                                                          | 3                                                                                               |
|  | 325                                                                                             | Emily Vargas                                                                             | 4                                                                                               |
|  | 376                                                                                             | Justin Donaldson                                                                         | 2                                                                                               |
|  | 486                                                                                             | Melanie Kelly                                                                            | 2                                                                                               |
|  | 461                                                                                             | Crystal Pierce                                                                           | 3                                                                                               |
|  | 2                                                                                               | Crystal Clements                                                                         | 2                                                                                               |
|  | 149                                                                                             | Jason Robinson                                                                           | 3                                                                                               |
|  | 173                                                                                             | Victoria Dixon                                                                           | 2                                                                                               |

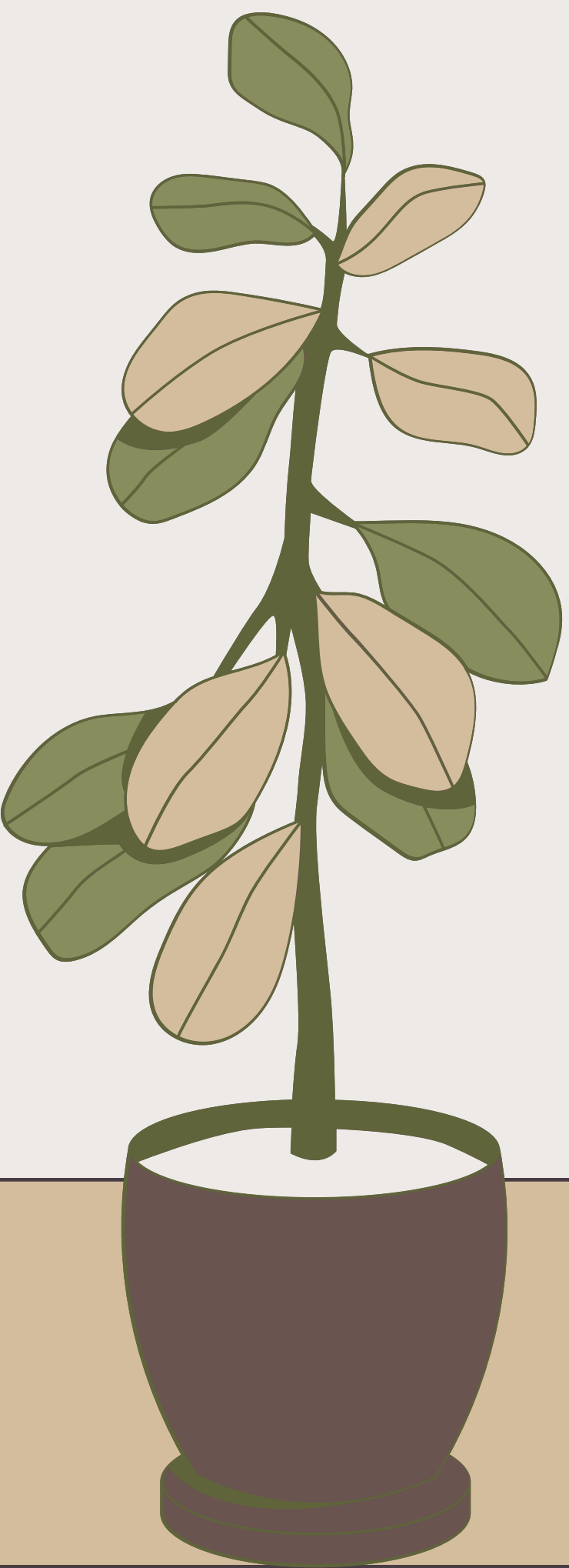




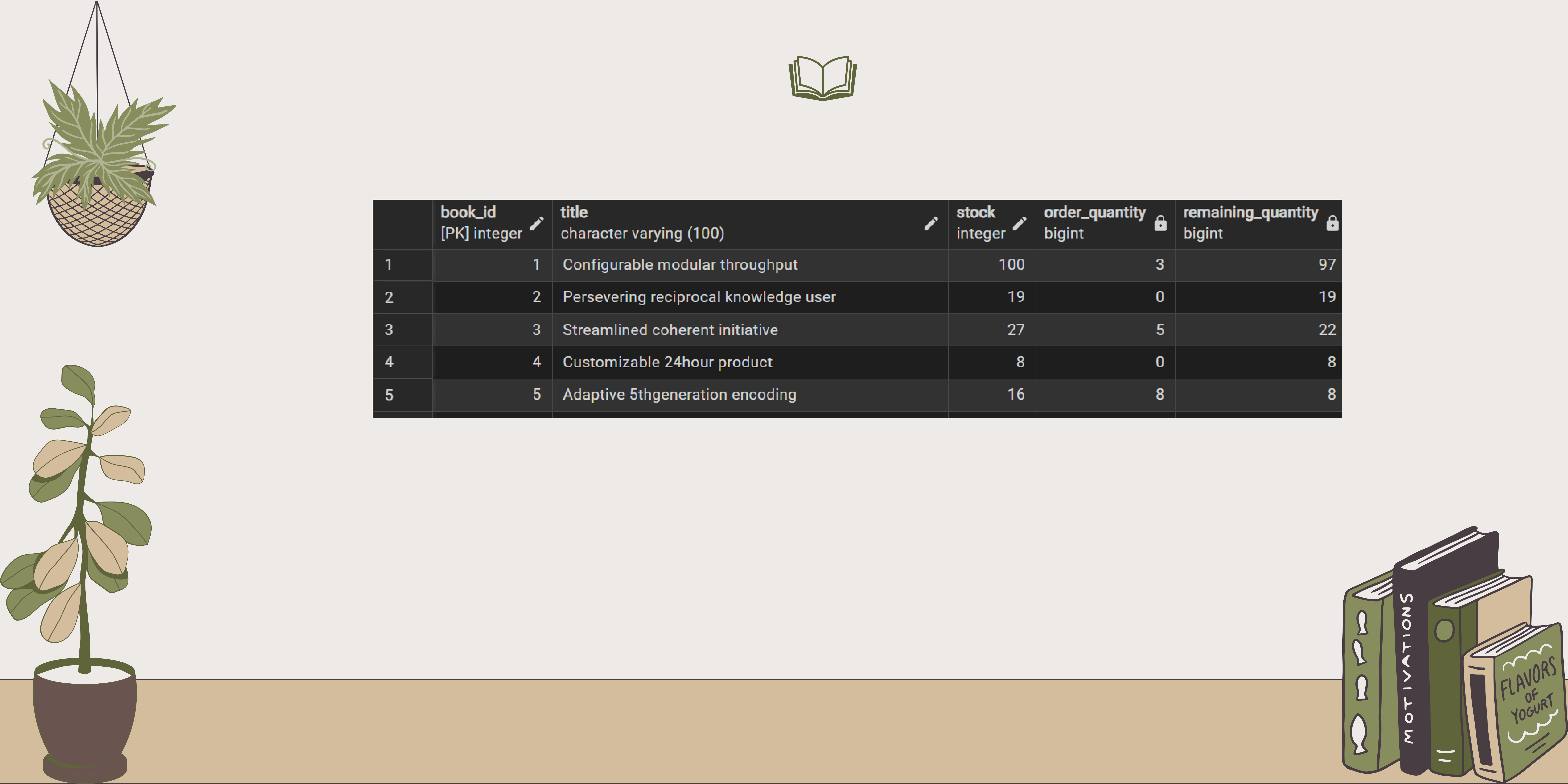


--Calculate the stock remaining after fulfilling all orders:

```
SELECT
  B.BOOK_ID,
  B.TITLE,
  B.STOCK,
  COALESCE(SUM(O.QUANTITY), 0) AS ORDER_QUANTITY,
  B.STOCK - COALESCE(SUM(O.QUANTITY), 0) AS REMAINING_QUANTITY
FROM
  BOOKS B
  LEFT JOIN ORDERS O ON B.BOOK_ID = O.BOOK_ID
GROUP BY
  B.BOOK_ID
ORDER BY
  B.BOOK_ID;
```







|   | book_id<br>[PK] integer  | title<br>character varying (100)  | stock<br>integer  | order_quantity<br>bigint  | remaining_quantity<br>bigint  |
|---|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1 | 1                                                                                                           | Configurable modular throughput                                                                                      | 100                                                                                                  | 3                                                                                                            | 97                                                                                                               |
| 2 | 2                                                                                                           | Persevering reciprocal knowledge user                                                                                | 19                                                                                                   | 0                                                                                                            | 19                                                                                                               |
| 3 | 3                                                                                                           | Streamlined coherent initiative                                                                                      | 27                                                                                                   | 5                                                                                                            | 22                                                                                                               |
| 4 | 4                                                                                                           | Customizable 24hour product                                                                                          | 8                                                                                                    | 0                                                                                                            | 8                                                                                                                |
| 5 | 5                                                                                                           | Adaptive 5thgeneration encoding                                                                                      | 16                                                                                                   | 8                                                                                                            | 8                                                                                                                |
|   |                                                                                                             |                                                                                                                      |                                                                                                      |                                                                                                              |                                                                                                                  |





# Thank You

